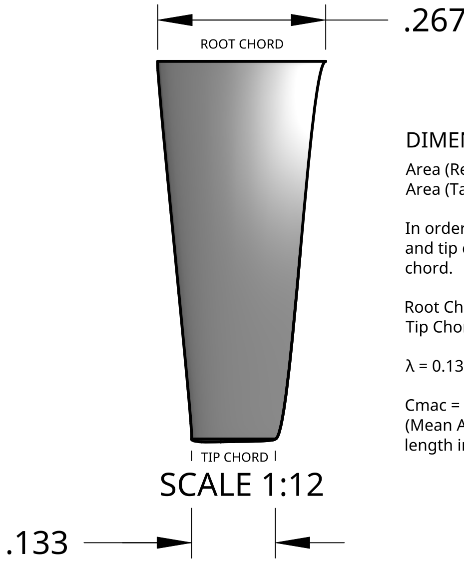
ROOT CHORD

TOP PLANFORM VIEW



SCALE 1:12

DIMENSION JUSTIFICATION

Area (Rectangular Wing) = 0.6 * 0.2
Area (Tapered) = 0.6 * [(Root+Tip)/2]

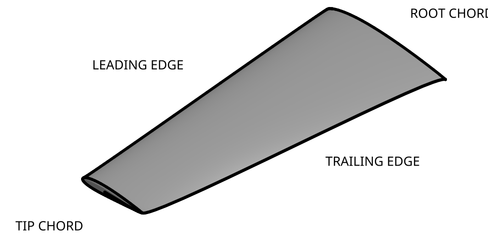
In order to keep span area constant, root chord and tip chord must average to the 200 mm base chord.

Root Chord = 0.2 (4/3) = 0.267
Tip Chord = 0.2 (2/3) = 0.133

λ = 0.133/0.267 = 0.498 = 0.5 (Taper Ratio)

Cmac = 2/3(0.267)(1+λ+λ^2/1+λ) = 0.2077
(Mean Aerodynamic Chord - used as reference length in control directory of OpenFOAM)

ISOMETRIC VIEW

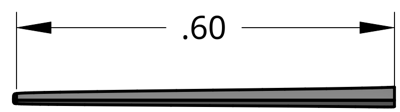


SCALE 1:12

SIDE VIEW

SCALE 1:12

LE/TE VIEW



SCALE 1:12

DIMENSIONS ARE IN METERS UNLESS MENTIONED OTHERWISE ROOT CHORD = 0.267 TIP CHORD = 0.133 TAPER RATIO = 0.5 SPAN = 0.6		NAME	DATE
	DRAWN	SPONDON SAHA	08/24/2026
	PROJECT	Comparative Analysis Of Wing Planforms	
DO NOT ALTER DRAWING	DRAWING IS MEANT TO BE USED IN PROJECT DOCUMENTATION OR FOR OTHER REFERENCES ONLY.		
TAPERED WITH RESPECT TO BOTH TE AND LE. MID-CHORD POINT REFERENCED.			
ROOT & TIP CHORD DIMENSIONS WERE CHOOSEN TO ENSURE CONSTANT SPAN AREA			

UNIFORM TAPERED WING

SCALE: MENTIONED IN DRAWING